Tech Products & Brand Perception

Data Intelligence and Sentiment Analysis on the Voice of the Consumer

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Module I: Context, Data Sources, and System Architecture

Business Motivation: Beyond the Code

- **The Core Business Problem:** Consumer electronics brands live and die by their public reputation. Conversations are scattered across highly unstructured channels.
- **Strategic Business Needs:** Marketing and product managers need to answer critical questions without looking at database consoles or raw logs:
  - How is brand sentiment changing week-over-week after a launch?
  - Which specific features (camera, battery, price) drive satisfaction or friction?
  - What is the difference between emotional "social buzz" and technical reviews?
- **The Proposed Solution:** A containerized, automated data intelligence pipeline that aggregates conversational sentiment with specialized user reviews into a unified reporting interface.

Data Sources: Real-time Buzz vs. Structured Reviews

Dimension	X / Twitter (Social Buzz)	GSMArena (User Reviews)
Nature	Spontaneous, emotional, short	Evaluative, technical, long-form
Ingestion Freq.	Twice a week (high temporal volatility)	Daily (slower changes, ethical delays)
Data Quality	Highly noisy (URLs, spam, bot accounts)	Moderate (long tech reviews, specs)
Supporting Metrics	Engagement (likes, reposts, views)	Numeric product rating (1-10)
Business Value	Real-time launch feedback, crisis tracking	Hardware failures, long-term degradation

Table 1: Strategic comparison of the pipeline's two primary data sources.

User Stories and System Boundaries

Core Functional User Stories

1. **Temporal Sentiment Trends:** Trace brand sentiment over time to measure campaign impact.
2. **Brand Benchmarking:** Compare brand metrics (e.g., Apple vs. Samsung vs. OnePlus).
3. **Friction Discovery:** Map specific hardware parts to negative comments.
4. **Anomaly Detection:** Spot sudden spikes in discussion volume or engagement.

Scope and Boundaries

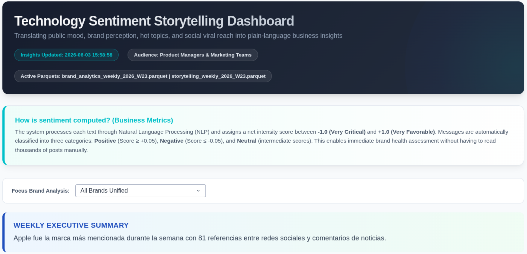
- **Language:** Restricted to English content to ensure VADER algorithm validity.
- **API Constraints:** RapidAPI limits daily request quotas on the free tier.
- **Freshness:** Scheduled batch ingestion (not streaming); weekly updates for curated Gold tables.

Data Pipeline: The Medallion Architecture

Logical Architecture Layers:

1. **Bronze (Raw Storage):** Unmodified JSON files directly from the APIs and scrapers to ensure full auditability.
2. **Silver (Cleaned):** Structured Parquet files with unified schemas, UTC normalization, and cleaned text.
3. **Gold (Curated):** Aggregated business KPIs and governance views ready for visualization.

Orchestrated by **Apache Airflow** in a containerized environment (Docker Compose) serving two Plotly Dash dashboards.



Logical architecture and components of the data pipeline.

Module II: Data Processing, Quality, and Governance

Data Purification: From Bronze to Silver

- **Data Standardization:** Timestamps standardized to UTC; data types converted (interaction counters mapped to integers).
- **NLP Preprocessing (Text Hygiene):**
  - Normalizing all characters to lowercase.
  - Stripping URLs and web links using regular expressions.
  - Removing username handles (‘@username‘) while extracting raw hashtags for context.
  - Cleaning residual HTML, special symbols, and punctuation.
  - Removing high-frequency stop words while preserving tech terms.
- **Initial Quality Filter:** Dropping comments under 20 characters to eliminate low-meaning filler (“ok”, “nice”, “yes”).

Resolving Critical Data Quality Issues

Handling Null Values

- If the ‘createdAt‘ timestamp or ‘text‘ is missing, the record is **dropped** immediately (no analytical value).
- If interaction metrics (likes, replies) are null, they are imputed to **zero**.
- Missing authors are imputed as “anonymous”.

Engagement Outliers

- Highly viral tweets skew arithmetic means, hiding general consumer behavior.
- We apply the Interquartile Range (**IQR**) method:
$$\text{IQR} = Q_3 - Q_1$$
- Outliers are flagged and capped at the **99th percentile** in storytelling tables to prevent visual graph distortions.

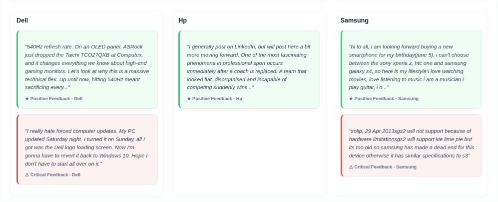
The Gold Layer: Batch Processing with PySpark

- **Why PySpark:** Enables scale-out computations. It consolidates raw temporal metrics and processes sentiment matrices in a distributed environment.
- **Airflow Orchestration:**
  - A **Triggers** monitors the Silver directory for new parquet files.
  - Spark transformations are triggered automatically every Monday at 2:00 AM.
  - Output is separated into two Gold subfolders: ‘governance/‘ (pipeline health logs) and ‘analytics/‘ (business storytelling).

Data Governance: Pipeline Health Indicators

We enforce pipeline health using automated quality KPIs:

1. **Null Rate:** Warns if column nulls exceed 5% (e.g., date columns).
2. **Duplicate Rate:** Warns if duplicate records exceed 2% (prevents duplicate API polling errors).
3. **Schema Compliance:** Ensures columns match specifications ($> 95\%$).
4. **Non-English Content Rate:** Monitors foreign languages to signal if translation engines are needed.



Null percentage monitoring on critical analytical columns.

Governance Dashboard (Port 8050)

Interface of the Governance Dashboard showing pipeline metrics.

Designed for Engineers and DBAs:

- Traffic-light state cards (Green / Yellow / Red) for instant quality checks.
- Historic volume charts tracking record ingestion per source.
- Schema validation checks to detect upstream source schema drift.
- Fully searchable data tables listing exact KPI values per weekly batch.

Module III: Analytical Storytelling and Business Insights

Sentiment Classification with VADER

- **Algorithm:** *Valence Aware Dictionary and sEntiment Reasoner*. Specifically tuned for social media (supports slang, capitalization, emojis, and punctuation modifiers).
- **Compound Score (S):**

$$S = \frac{\sum V_i}{\sqrt{(\sum V_i)^2 + \alpha}} \quad \text{where } \alpha = 15$$

- **Custom Sentiment Boundaries:**
  - **Positive:** $S > 0.2$ (clear satisfaction and recommendation)
  - **Negative:** $S < -0.2$ (criticisms, hardware issues, cost complaints)
  - **Neutral:** $-0.2 \leq S \leq 0.2$ (objective specifications, pricing inquiries)

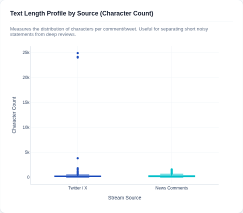
Consolidated Sentiment Distribution

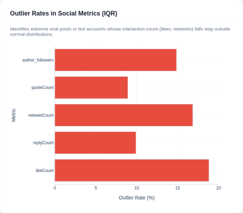
Sentiment Breakdown by Channel

- **X / Twitter (Polarized):** High positive engagement (45%) and marked negative comments (20%). Low neutral proportion (35%).
- **GSMArena (Analytical):** Higher neutral percentage (42%) due to user focus on comparing technical specifications.
- **Combined Average:**
  - 42% Positive
  - 38% Neutral
  - 20% Negative

Volume distribution by dataset and classification categories.

Key Perception Drivers (Keyword-Sentiment Matrix)





Word frequency of product attributes in cleaned datasets.

Mapping keywords against average sentiment scores highlights product strengths and weaknesses:

- `camera` (Mean Sentiment: **+0.45**): The primary driver of satisfaction (low-light performance, optical zoom).
- `display` (Mean Sentiment: **+0.28**): Praise for 120Hz refresh rates and OLED color accuracy.
- `battery` (Mean Sentiment: **+0.12** on Twitter, **-0.05** on GSMArena): Polarized. Fast charging is loved, but long-term battery degradation is heavily criticized.
- `price` (Mean Sentiment: **-0.35**): Major source of criticism. High premium brand costs reduce overall value perception.

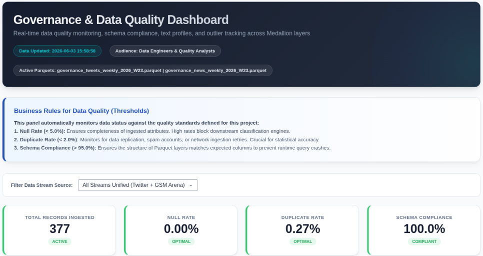
Brand Perception Comparisons

- **Apple:** Highest conversation volume. Extreme polarization. High positive scores for cameras, offset by severe negative sentiment regarding battery replacement costs and initial price.
- **Samsung:** Steady discussion on technical specifications. High satisfaction regarding screen and folding form factors (Galaxy Fold/Flip), but criticisms focus on slow charging innovations.
- **OnePlus:** Slower mention rate, but displays the highest net sentiment score. Praised for its cost-to-performance ratio and Warp Charge capabilities.

The Virality Phenomenon on Social Media

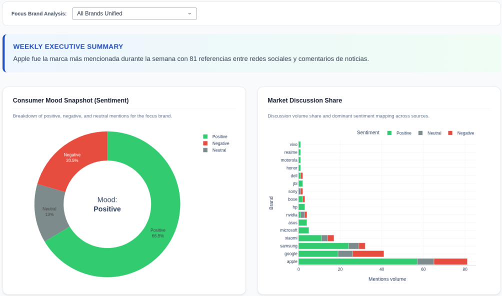
Engagement Metrics vs. Sentiment Polarity

- By plotting sentiment scores against engagement counts (likes, retweets, views), we identified a critical trend.
- **Key Finding:** Highly viral tweets (extreme engagement outliers) tend to skew heavily toward negative sentiment or sarcasm.
- **Business Lesson:** Marketing departments should not react blindly to individual viral posts. Sarcastic posts capture clicks, but a balanced view requires looking at stable forum reviews.



Boxplot showing engagement outlier distributions.

Storytelling Dashboard (Port 8051)



View of the Storytelling Dashboard designed for marketing analysts.

Designed for Fast Marketing Insights:

- Auto-generated plain-language narrative summarizing weekly metrics.
- Sentiment donut chart showing general brand health.
- Brand ranking comparison by conversation volume.
- Interactive Quote Cards: Renders the most positive and most critical raw comments for the selected brand, allowing rapid qualitative verification.

Module IV: Technical Milestones, Lessons, and Future Work

Technical Milestones and Implementation Challenges

1. **Parquet Timestamp Incompatibility:** Pandas wrote timestamps in nanoseconds (TIMESTAMP_NANOS) which Spark rejected natively. Resolved by designing a dynamic fallback reader checking schemas.
2. **Worker UDF Serialization:** Inline Spark functions in Airflow tasks raised serialization errors on distributed workers. Resolved by moving transformations to ‘gold_helpers.py‘ and distributing it via ‘addPyFile()‘.
3. **Docker Volume Writing Permissions:** The Airflow Celery worker running as UID 50000 lacked permission to write output Parquet directories on the host. Solved by injecting owner chown mapping into the compose startup scripting.

Conclusions and Enterprise Next Steps

Key Conclusions

- The project successfully builds a bridge from messy online comments to clean, structured business insights.
- Automated quality gates (Governance) are crucial; without them, business dashboards report errors silently.
- Balancing high-speed social API data with slow reviews limits viral noise.

Recommended Next Steps

- **Real-Time Stream Processing:** Integrate Apache Kafka and Spark Streaming to reduce latency from days to seconds.
- **Aspect-Based Sentiment Analysis:** Replace the lexicon model (VADER) with fine-tuned Transformers (**RoBERTa** or LLMs) to link sentiments to specific hardware components.
- **Cloud Dataproc Migration:** Transition from the local Docker stack to cloud-native managed Spark and Airflow.

Questions & Answers